

Design Calculation Sheet

Parameter	Value	Unit	Formula/Source
Nominal capacity	3.28	Ah	Simulation: r5_Q_1c.json
1C current	3.28	A	= Nominal capacity × 1
Discharge energy	11.84	Wh	$= \int V \cdot I \, dt / 3600$ (trapezoidal)
Cell mass (excl. electrolyte)	33.4	g	= Σ layer thickness × (1- ϵ) × density × area
Energy density (gravimetric)	378.69	Wh/kg	= 11.84 Wh / 0.0334 kg
Midpoint voltage	3.61	V	= Voltage at discharge midpoint
DC resistance	0.35	m Ω	= (OCV_start - V_10%) / I_1C
Power density	269.7	W/kg	= V_OC ² / (4 · DCR) / mass
Positive electrode thickness	40	μ m	Parameter override
Negative electrode thickness	60	μ m	Parameter override
N/P ratio	1.5	—	= 60/40
Positive particle radius	2	μ m	Parameter override
Negative particle radius	5	μ m	Parameter override
Electrolyte diffusivity	2.0e-9	m ² /s	Parameter override (estimate)
Electrolyte conductivity	2.5	S/m	Parameter override (estimate)
Cation transference number	0.38	—	Parameter override (estimate)
SEI kinetic rate constant	7e-15	m/s	Parameter override (estimate)
4C charge T_max	344.95	K	Simulation: r5_Q_4c.json
4C min anode potential	0.0006	V	Simulation: r5_Q_4c.json
Low-T (-20°C) retention	99.99	%	= cap_lowT / cap_1C × 100
SEI thickness @100 cyc	205.9	nm	Simulation: r5_Q_aging100.json
SEI thickness @500 cyc	518.1	nm	Simulation: r5_Q_aging500.json